

Functionality & Business Logic in Java Programming

1. Introduction

In Java programming, functionality refers to the specific tasks or operations that a program performs.

2. What is Business Logic?

Business logic is the custom rules or algorithms that handle the exchange of information between a database and service layer.

3. Separation of Concerns

Separating business logic from other layers like UI and data access improves maintainability and scalability.

4. . Example of Business Logic in Java

Ex:

```
public class OrderService
{
    public double calculateTotal(double price, int quantity) {
        if (quantity <= 0) throw new IllegalArgumentException("Quantity must be positive");
        return price * quantity;
    }
}
```

5. Best Practices

- Keep business logic in service classes, not in controllers or models.
- Use meaningful method names that reflect business operations.
- Write unit tests for business logic.
- Avoid hardcoding values; use constants or configuration files.
- Document business rules clearly in code comments or external documentation.